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Topic: Data Cleaning using Power Query

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Data Cleaning using Power Query

1. Problem Statement

A company has received customer transaction data containing missing values, duplicate records, and inconsistent entries. The objective of this assignment is to clean and prepare the dataset using Microsoft Power Query so that it is accurate, consistent, and ready for further analysis.

2. Dataset Overview

The dataset used for this exercise, named `dirty_cafe_sales`, contains 1,000 transaction records from a café, spanning the following seven columns: Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, and Location. In its raw form the dataset contained blank cells, inconsistent placeholder text, duplicate rows, and untrimmed text entries. A sample of the cleaned data is shown below.

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
TXN_1961373	Coffee	2	2	4	Credit Card	Takeaway
TXN_4977031	Cake	4	3	12	Cash	In-store
TXN_3160411	Coffee	2	2	4	Digital Wallet	In-store

Table 1: Sample records from the cleaned `dirty_cafe_sales` dataset

3. Methodology

The dataset was imported into Power Editor / Power Query within Power BI and cleaned through a sequence of applied steps. Each step addressed a specific data quality issue — inconsistent headers and data types, invalid or placeholder values, missing rows, duplicate transactions, and untrimmed or unprintable text. The applied steps, in the order they were executed, are summarised in Table 2.

#	Applied Step	Purpose
1	Source	Connected Power Query to the raw <code>dirty_cafe_sales</code> dataset containing 1,000 transaction records.
2	Changed Type	Set an initial data type for each column as detected on import.
3	Promoted Headers	Used the first row of the source data as column headers (Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location).
4	Changed Type 1	Re-applied correct data types after promoting headers — text for identifiers, whole/decimal numbers for Quantity, Price Per Unit and Total Spent.

5	Replaced Value / Replaced Value 1	Replaced inconsistent and placeholder entries (e.g. blank, "ERROR" or "UNKNOWN" text) in the affected columns with appropriate null or standard values.
6	Changed Type 2	Corrected data types again after the replacements so numeric columns remained numeric and text columns remained text.
7	Filtered Rows	Removed rows with missing or invalid Transaction ID / Quantity values that could not be reliably corrected.
8	Removed Duplicates	Removed fully duplicated transaction records to ensure each row represents a unique sale.
9	Trimmed Text	Removed leading and trailing spaces from all text columns (Transaction ID, Item, Payment Method, Location).
10	Cleaned Text	Applied Text.Clean to strip non-printable characters from text columns, using Table.TransformColumns on the "Transaction ID", "Item" and "Payment Method" fields.

Table 2: Applied steps used to clean the dataset in Power Query

3.1 Handling Missing Values

Blank and placeholder entries in numeric and text columns were identified using the column quality indicators in Power Query and replaced or removed using the Replace Values step so that no column retained invalid placeholder text.

3.2 Removing Duplicate Records

The Remove Duplicates step was applied across all columns to ensure that each transaction in the dataset was represented only once, preventing double counting in later analysis.

3.3 Standardising Inconsistent Entries

Text columns such as Transaction ID, Item, and Payment Method were standardised using the Trim and Clean transformations. The final Cleaned Text step applied the Text.Clean function through Table.TransformColumns, as shown in the formula bar below, to remove non-printable characters from the key text columns.

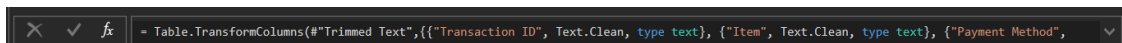


Figure 1: Power Query formula bar showing the Table.TransformColumns step used for the Cleaned Text transformation

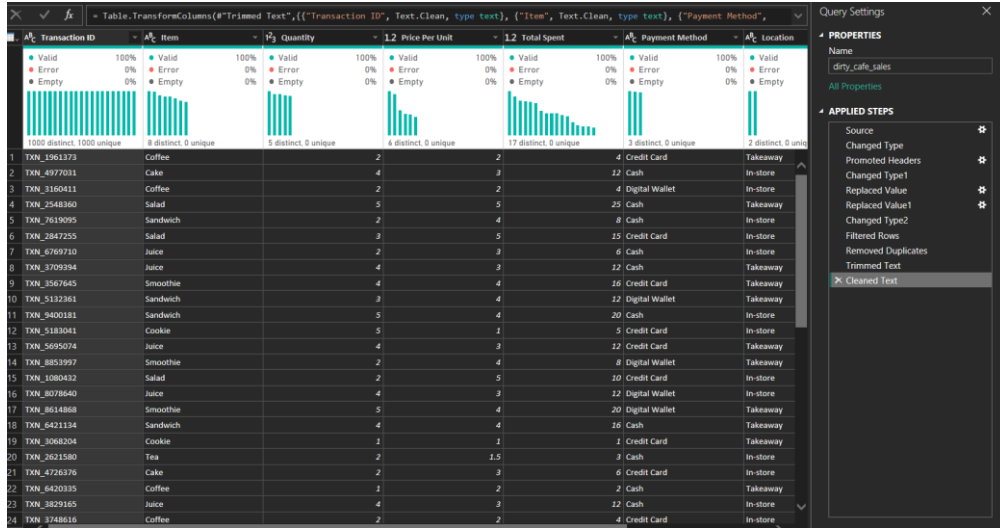


Figure 2: Power Query Editor showing the complete list of Applied Steps for the *dirty_cafe_sales* query

4. Results

After all cleaning steps were applied, the column quality indicators in Power Query confirmed that every column in the dataset reached 100% valid values, with 0% errors and 0% empty cells. This confirms that the missing values, duplicate records, and inconsistent entries originally present in the dataset were successfully resolved. A summary of the column-level data quality after cleaning is shown in Table 3, and the corresponding Power Query column-quality view is shown in Figure 3.

Column	Valid %	Error %	Empty %	Distinct Values
Transaction ID	100%	0%	0%	1000 distinct, 1000 unique
Item	100%	0%	0%	8 distinct, 0 unique
Quantity	100%	0%	0%	5 distinct, 0 unique
Price Per Unit	100%	0%	0%	6 distinct, 0 unique
Total Spent	100%	0%	0%	17 distinct, 0 unique
Payment Method	100%	0%	0%	3 distinct, 0 unique
Location	100%	0%	0%	2 distinct, 0 unique

Table 3: Column-level data quality summary after cleaning

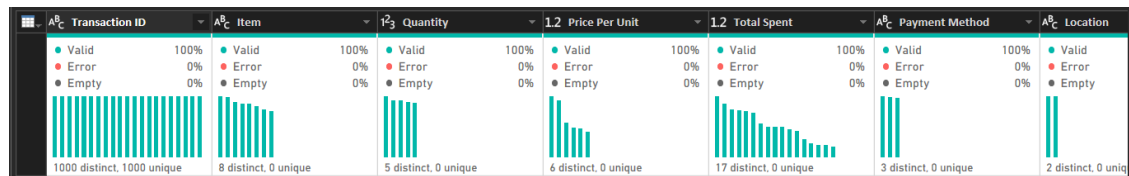


Figure 3: Power Query column quality indicators after cleaning — all columns show 100% valid values

5. Conclusion

Using Power Query, the raw cafe sales dataset was systematically cleaned through header promotion, data type correction, value replacement, filtering, duplicate removal, and text trimming/cleaning. The resulting dataset of 1,000 unique, fully valid transaction records is now consistent and reliable, and is suitable for further analysis and reporting in Power BI.